

EQUESTRIAN WASTELAND SIM (EWS) — MASTER AI HANDOFF & PROJECT BIBLE

*Document Version: 2.0.0 | Date: 2026-10-06 | Author: Dylan Weaver & Gemini Spark Assistant
Target Audience: Any AI Model (Claude, ChatGPT, Gemini) or Human Developer continuing this project.*

1. OWNER & WORKFLOW PROFILE

- **Project Owner:** Dylan Weaver (illustrator, character designer, non-programmer, voice-dictates on iPad Air, Missouri, US).
 - **Core Directive:** Dylan thinks and designs in visual shape language, charm, and game feel. Never speak down to him with abstract technical jargon; explain mechanics and fixes plainly in terms of visual and gameplay results.
 - **Workflow Environment:** iPad Air (2026), Apple Pencil & touch-first, Textastic Code Editor (local web server at <http://localhost:8080/>), Safari WebKit / Home Screen PWA.
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2. ART STYLE & SHAPE LANGUAGE PHILOSOPHY

- **The Anti-Photorealism Law:**
 - Strictly avoid raw photographic textures, realistic skin pores, photorealistic PBR noise, or complex micro-strand hair (NO 400 loose hair strands!).
 - The game is an expressive, stylized, charming toon world. Shapes must be bold, readable, and clean with flat colors, soft toon lighting, and harmonizing gradients.
 - The beauty of the game comes from pure mathematical volume and silhouette—a modular, lightweight character that runs at 60 FPS on mobile with a fraction of the compute power of typical 3D rigs.
- **Two-Piece Modular Mane Architecture:**
 - Pony manes must always be built from two distinct, modular anatomical pieces:
 1. **Nape / Crest Mane (Back Neck Hair):** Anchors along the dorsal ridge of the neck from the poll down to the withers. Depth-layered behind the skull/face, but in front of the withers and back body. Can be long flowing locks (Rainbow Dash style), short tomboy spikes (Scootaloo style), or a sculpted arching crest.

2. **Forelock / Crown Mane (Brow & Bangs):** Sits on the upper cranium and forehead, framing the eyes, brow, and temples.
 3. **Tail:** A volumetric S-curve plume anchored at the sacral dock on the pelvis.
- Constructed using chunky, tapered volumetric ribbons (thick root tapering to soft rounded tips) rather than thin wire strands.

3. MASTER GAMEPLAY & LORE BIBLE

- **Setting:** Post-apocalyptic wasteland simulation set in Equestria, presented in a 2.5D isometric tile world (64x32 isometric projection) with real-time procedural 3D characters.
 - **Equine Factions & Sociology:**
 - **Settler & Vault Ponies:** Civilized, technological, highly structured. They value cleanliness, waste containment, and indoor sanitation facilities.
 - **Wild & Feral Wasteland Ponies:** Roam the wastes, un-housebroken, eliminating freely outdoors in dirt, mud, and water ("wild ponies are dirty and pee"). They require socialization, taming, or acclimation to vault living.
 - **Childcare & Foal Mechanics:**
 - Diapers exist as early childcare equipment for un-trained foals.
 - Potty-training skill progression: Level 0 (Untrained, requires diapers or soils surroundings), Level 1 (Learning / rudimentary), Level 2 (Fully housebroken).
 - Sanitation & hygiene dissatisfaction penalties apply if foals or surroundings are left soiled.
 - **Equine Internal Biology & Genetics:**
 - **9-Point Authentic Hindgut Fermentation Spline:** Functional Three.js Catmull-Rom spline linking mouth -> esophagus -> stomach vat -> cecum fermentation chamber -> large colon loop -> caudal aperture ring.
 - **Fermentation Gas Dynamics:** The cecum accumulates gas pressure from microbial fermentation (0–100%). High pressure causes visible flank distension and requires periodic venting.
 - **Contraband Ingestion & Smuggling Pipeline:** Ingested non-digestible contraband (Bobby pin lockpicks, jailbreak skeleton keys, encrypted holo-chips) progresses along the digestive tract to the exit aperture for extraction.
 - **Mendelian Coat Genetics:** Punnett square inheritance engine modeling Extension (\$E/e\$), Agouti (\$A/a\$), and Cream dilution (\$Cr/n\$) alleles to produce authentic equine coat phenotypes (Bay, Chestnut, Palomino, Buckskin, Cremello, Perlino, and Raven Black).
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4. MASTER USER INTERFACE SPECIFICATION: THE ZERO-BAR LAYOUT LAW

Crucial: If any AI asks about the UI, this is the authoritative specification.

- **Zero-Bar Principle:** Exactly 0 persistent top or bottom chrome bars. 100% of the screen is a borderless 3D viewport.
- **Outer Margin (--m = 12px):** Reserved strictly for circular frosted glass orbs (44px touch targets):
 - Bottom-Left (#mode): Simulation Pause / Live toggle.
 - Bottom-Right (#time): Simulation Speed stepper (1x, 2x, 5x).
 - Top-Right (#sub): Cinematic low-angle ground camera toggle.
 - Bottom-Left Upper (#rag): Active Ragdoll physics toggle (Off, Standing Physics, Free Fall).
 - Bottom-Left Mid (#bld): Stage 3 Build Mode editor overlay toggle.
 - Top-Left (#layer-dropdown): Glass dropdown for hierarchical layer isolation.
- **Inner Margin (--mi = 60px):**
 - **Draggable Revolver 6-Shooter UI:** 150x150px touch dial with central clock/state readout and 6 radial bullet buttons. Auto-maps contextually between **LIVE** mode (Cutaway, Ragdoll, Fluids, Contraband, Genetics, Camera, Sync), **BUILD** mode (Gizmo, +Ball, +Hole, Mirror, Slice, Presets), and **SELECT** mode (Inspect, Dissect, Rig, OBJ Export). Can be dragged anywhere on screen and snaps within safe margins.
 - **Sliding Glass Sheets (#sheet):** Slide up from the bottom on-demand when summoned from the Revolver or orbs. Fully touch-scrollable with 44px buttons and sliders. Automatically dismisses upon tapping empty ground.
- **Hierarchical Layer Grouping & Isolation:**
 - Layer 5: Armor & Outer Shell
 - Layer 4: Clothing & Wearables
 - Layer 3: Surface Skin & Coat
 - Layer 2: Skeletal Rig & Bones (3D animated bone lines + joint diamond markers)
 - Layer 1: Deep Viscera & Organs (stomach, cecum, colon, bladder, repro tract)
 - **Nude Base View:** Clean, smooth chibi pony form with solid connected anatomy, organs and skeleton completely hidden inside, no clothing or armor.
- **Action & Behavioral Director:**
 - Initiator -> Target Pairing: Me -> Cindy, Cindy -> Charlie, Self -> Mirror.
 - Actions: Hug, Kiss / Nuzzle, Salute, Groom, Brawl.
 - Intensity Stepper: 25% Soft, 50% Mid, 75% Firm, 100% Fierce (controls active ragdoll spring tension).

- Relationship Loops: Once, Doting / Frequent, Obsessive Loop (e.g. self-absorbed mirror kissing loop).
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5. 3D TECHNICAL ARCHITECTURE & MESH ENGINE

- **Primary Game Client:** `EWS_with_Build.html` (single-file self-contained Three.js r128 application).
 - **Core Math Rig (MYP in `pony_geometry_rig_math.js`):**
 - 23-node canonical sphere model authored by Dylan (`head`, `muzzle`, `jaw`, `cheekL`, `cheekR`, `chest`, `rear`, `pelvis`, `FL1..FL4`, `FR1..FR4`, `BL0..BL4`, `BR0..BR4`).
 - **Solid Body Connectors (`bodyConnectors`):**
 - Closed, double-sided cylinders (`THREE.DoubleSide`, `openEnded: false`) bridging every joint:
 - Torso barrel connecting `rear` to `chest`.
 - Neck connecting `head` to `chest`.
 - Muzzle & jaw connecting to `head`.
 - All four leg chains (`FL`, `FR`, `BL`, `BR`).
 - Updated every frame in `animate()` to stretch, orient, and match walk gait and ragdoll physics seamlessly. Completely eliminates "floating bead" monsters.
 - **Build Mode Raymarching (`st3d_viewport.html`):**
 - Shader-based SDF raymarcher embedded via base64 `ST_SRC` in an iframe.
 - **Anatomical Hard Union:** Separates shapes into Trunk ($|y| \leq 0.05$), Left Limbs ($y > 0.05$), and Right Limbs ($y < -0.05$). Left and right limbs smooth-blend into the torso trunk, but are hard-unioned with each other using $\min(d_L, d_R)$. Left and right legs NEVER melt across the midline.
 - **Direct Radius Halo:** Dotted perimeter ring around selected spheres allows direct touch-drag radius adjustment without opening sliders.
 - **3D Opening & Cavity Mapping:** Cut nodes are mapped in 3D: `hole` generates a caudal aperture torus ring snapped to the digestive spline exit; `bag` generates an internal cavity shell.
 - **Locomotion & Touch Physics:**
 - **Four-Beat Horse Walk Gait:** Lateral sequence (BL, FL, BR, FR, 25% phase offset, ~62% stance duty, natural hock/knee folding).
 - **Position-Based Dynamics Ragdoll:** Verlet physics with environmental avoidance vectors (pushes away from toxic acid pools, waste grates, and ruins).
 - **iPad Touch Disambiguation:** 1-finger brush/drag = steering locomotion; 2-finger drag = camera orbit; 2-finger pinch = camera zoom; tap on character = open cutaway; tap on empty ground = deselect / close sheets.
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6. MASTER DIRECTORY & CLOUD VAULT MAP

All files reside in Google Drive under Dylan's master vault root:

[1kfs1MRCdE6_jzMUEijTD7JN6FfTp0_8d](#).

Folder / Path	Drive Folder ID	Scope & Purpose
Master Vault Root	1kfs1MRCdE6_jzMUEijTD7JN6FfTp0_8d	Root container for the live game client (EWS_with_Build.html), Master AI Handoff, and Project Bible.
tools/	10mXWrdNKt71SFNW1G3CP0pMUmVph5bY5	Standalone dev tools: st3d_viewport.html (Stage 3 3D Build Viewport), echo.py (state & heartbeat logger), and pipeline scripts.
Rigs_and_Art/	1TT4P335EF36f2CPhdksia81BYL50tpWX	Geometry math and schemas: pony_geometry_rig_math.js (canonical MYP 23-node math), pony_blockout_v2.json , pony_stallion_v2.json .
Saves/	1zTK15kWxS6Kope6mt6g3Z-RNw4xDUK2H	Game saves, character states, and exported blockout JSONs.
Mods_and_Lore/	15m1MIFWdNnTcVJBw6wYxs01MpTKfb8dT	Story bible, wasteland lore, dialogue scripts, and EWS_GameMaster_Schema.json (traits, perks, weapons, armor, consumables).
Backups/	1Hqqv3ttckz dqXrgbqzTGQdCTPX4c9vRc	Full milestone archives and zip bundles (e.g., ST3D_BUNDLE_11.zip).

7. IPAD APP LAUNCHER & TEXTASTIC QUICK REFERENCE

1. **Running Fullscreen on iPad:** Open `EWS_with_Build.html` in Safari → Tap Share button → Tap **"Add to Home Screen"**. The game launches as an official borderless standalone web app with zero browser bars.
 2. **Textastic Local Workflow:** In Textastic, open the file and tap Preview. Textastic serves it locally at `http://localhost:8080/EWS_with_Build.html`. You can bookmark that URL in Safari or open it in iPad Split View.
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8. CURRENT LIVE DEBUGGING STATUS (OCTOBER 6, 2026)

3. **Reported Bug:** On iPad Air (2026) running Textastic web preview (`http://localhost:8080/`), the 3D viewport canvas (`#vp`) displays solid clear-color dark green/black (`#0c1410`). The HTML/CSS UI overlay (corner orbs, Revolver 6-shooter dial, sliding sheets) compiles and renders correctly. When tapped, sliding sheets (such as 3D Avatar & Breeds, Visceral Cutaway) open and sliders function, but the 3D meshes (terrain ground grid, 23-sphere pony, props) are not visibly rendering on the canvas.
 4. **Touch Interaction Issue:** The draggable Revolver dial is not responding to touch drag gestures, and screen touches are only registering on DOM text/buttons, suggesting touch events may be intercepted or pointer-capture listeners in the active ragdoll / steering modules are consuming pointer events prematurely.
 5. **Client File:** `EWS_with_Build.html` (single-file Three.js r128 web client).
 6. **Core Directive for Claude:** Review the rendering pipeline, camera projection/positioning, material clipping planes, and pointer/touch event routing to resolve the green screen and restore full touch interactivity without violating the Zero-Bar Layout Law or the Anti-Photorealism Shape Law.
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9. AI MATHEMATICAL CLIPBOARD BRIDGE & LIVE INJECTION ENGINE (ews.math/1)

7. **1-Tap Clipboard Export (`copyAIMathPacket()`):** In SELECT or MODEL mode, tapping  **Copy for AI** captures an analytical mathematical JSON payload of the targeted node, active visceral state (stomach, cecum, gas pressure, bladder, gestation), and all canonical 23-node local coordinates `[x, y, z]`, radii `r`, and scales `[sx, sy, sz]` directly to the iPad clipboard.
8. **Instant Live Injection (`injectAIMathPacket(json)`):** Tapping  **Inject AI Math** allows pasting mathematical delta patches generated by Gemini or Claude directly into the running game engine. It applies positional offsets, scale tweaks, or organ volume deltas live in-place without reloading, downloading, or file renaming.

9. **Camera Third-Person Chase Setup:** Initial camera starts directly behind the character rump facing +X (`camAngle = {yaw: -PI/2, pitch: 0.32, dist: 5.2}`), vertically tracking the pony center of mass at $y = 0.85$ to prevent head clipping.
 10. **Left-Side Blender Views Stack:** Collapsible accordion for Free Cam, Isometric, Front (+X), Side (+Z), Top (+Y), Wireframe, and X-Ray Ghost toggles.
 11. **Anatomical Hair & Tail Locking:** Crown mane parented to the head sphere, tail anchored to pelvis dock, preventing mid-air detachment during movement or ragdoll physics.
 12. **Organ Bag Centering:** `organGroup` positioned at local (0,0,0) inside the torso barrel, eliminating floating ghost viscera. Dynamic distension pushes the ventral belly mesh downward and laterally when organ volume exceeds 65L.
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10. CHUNK-BASED PROCEDURAL VOXEL EXCAVATION & PAINTABLE COLONY SPECIFICATION

13. **Master Engine Mode Stack:** 4 persistent primary states accessible at top-left: **SELECT** (🎯), **EDIT** (✏️), **BUILD** (🔨), and **ANIMATE** (🎬).
 14. **ANIMATE & Isolated Feature Staging:** Enables dynamic timeline scrubbing and module test stages (e.g. Cutaway Gas Room testbed with walled perimeter, open sagittal camera view, toxic mustard gas particle emitter, and live avoidance/coughing evaluation).
 15. **Procedural 3D Chunk Architecture (RimWorld/Dwarf Fortress Multi-Level Model):**
 - Horizontal 16×16 tile sectors (0.5m tile pitch) with discrete height levels (Z-levels from +3 Towers down to -2 Subterranean Vaults).
 - Memory-efficient flat typed array per chunk: `UInt8Array(16 * 16 * numZLevels)` (2KB RAM per chunk), ensuring 60 FPS on mobile iPad Air without WebKit memory bloat.
 - Surface face culling: Only exposed top and cutaway boundary faces are rendered.
 16. **Paintable Colony Designations:**
 - Drag-box and brush painting for **Mine / Dig** (amber glow) and **Stockpile** (cyan glow).
 - Autonomous Job Queue: Equines with hooves/tools pathfind to designated blocks, apply excavation damage ticks, drop resources (+1 Wasteland Mud/Stone), and haul materials to designated stockpiles.
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11. 200x200 MULTI-LEVEL WORLD SPACE, BASE BIOMES, AND HORIZON SKIRT ARCHITECTURE

17. **World Space Specifications:**

- **Playable Territory:** 200 × 200 tiles at 0.5m pitch = 100m × 100m active simulation space.
- **Verticality (Z-Levels):** Discrete vertical strata from -2 (Deep subterranean vault & bedrock) to +3 (Watchtowers & high battlements), with Level 0 as base ground.
- **Performance Guarantee:** Flat typed memory array `UInt8Array(16 * 16 * 6)` = 1.5 KB per chunk. The full 169-chunk world consumes under 250 KB of RAM, maintaining 60 FPS on iPad Air Safari.

18. Base Biomes & Toon Shape Language (Anti-Photorealism):

- **Borderland Woods & Streams (Starting Zone):** Lush moss-green turf (`#3a5a40`) and soft sage clover hills (`#588157`). Winding 3-tile wide stream channel (`#48cae4`) with rounded pebble banks. Clustered low-poly trees with chunky warm-brown trunks and tiered rounded sphere crowns.
- **Open Plains:** Sun-bleached prairie grass (`#606c38`) with wild golden oats (`#dda15e`). Shallow watering holes (`#64dfdf`) and solitary windswept landmark trees.
- **The Desolation Wastes:** Scorched dark earth (`#43281c`) with dry fissure crusts (`#7f4f24`). Dead calcified snag trees with angular branching silhouettes. Radioactive neon runoff pools (`#76ff66`).

19. Distant Horizon Skirt Math ("Bullshitting Unusable Distant Space"):

- **The Multi-Ring Illusion:** To prevent the world from feeling like a giant box or floating slab, an outer procedural mountain ring wraps around the 100m playable box (inner radius 52m, outer radius 650m, 48 radial segments, 192 polygons total).
- **Canterlot Ridge Silhouettes:** Low-poly mountain peaks (8m to 65m high) generated by sine wave harmonics, fading into atmospheric fog (`THREE.FogExp2`).
- **Zero Compute Overhead:** The horizon ring is static geometry sharing the master terrain shader, giving the illusion of an infinite post-apocalyptic continent without processing unusable distant tiles.

20. Colony Job Queue & Work Flow:

- **Designations:** `GameMaster.jobQueue` handles paintable mining (`addMineDesignation`) and storage allocations (`addStockpileTile`).
- **Worker Pipeline:** Idle equines or selected workers claim nearest designated jobs, pathfind across 2D tile grids with Z-level transitions, apply excavation damage ticks, drop loot (+1 Wasteland Mud/Stone), and haul resources to designated cyan stockpiles.

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12. DIRECT TOUCH VOXEL BUILDING & FACE-NORMAL SNAPPING SPECIFICATION

25. Direct Player Building vs. Equine Worker Digging:

- **Player Direct Build:** When the user enters **BUILD** mode and selects a block type, tapping any block face in the 3D world instantly snaps and commits the block into the world (no waiting for AI colony queues).
- **Equine Stat-Based Digging Ticks:** Digging a block is not instantaneous. An unequipped earth pony with baseline strength takes ~2.5 seconds (timed progress ring / cracking decal stages) to break a mud/dirt block. Equipping forged picks or having higher physical stats reduces dig duration.

26. Face-Normal Dynamic Snapping Logic:

- **Intersection Raycasting:** Tapping on a block retrieves `intersection.face.normal` directly from Three.js:
 - Top Normal (0, 1, 0): Snaps on top of the block (raising walls, laying floors, stacking pillars).
 - Bottom Normal (0, -1, 0): Snaps underneath the block (building ceilings, archways, and cave roofs).
 - Lateral Normals ($\pm 1, 0, 0$) or (0, 0, ± 1): Snaps to the side faces (extending walls, framing doorways).
- **Formula:** $P_{\text{new}} = P_{\text{target}} + (N_{\text{face}} \times \text{blockSize})$.

27. Touch-First Vertical Block Palette (Zero-Bar Law):

- **Placement:** Vertical frosted glass rail docked along the right inner margin (`--mi = 60px`), leaving 100% of the viewport visible.
- **Touch Chips ($\geq 44\text{px}$):**
 -  **Mud Block** (gathered from soil/riverbeds)
 -  **Stone Brick** (quarried limestone/granite)
 -  **Treated Timber** (hewn wasteland logs)
 -  **Vault Metal** (salvaged scrap plates)
 -  **Pick / Demolish** (targeted block removal)
- **Subtle 3D Grid Overlay:** Automatically renders a soft, faint grid along the active Z-level floor in **BUILD** mode to provide spatial grounding.

28. Two-Finger Camera Roll & Perspective Control:

- One-finger tap = place active block or select.
- Two-finger drag = orbit/tilt camera around the pony to view ceilings, underside overhangs, and rear corners effortlessly.
- Two-finger pinch = zoom in close for precision placement.

29. InstancedMesh GPU Architecture (60 FPS on iPad Air):

- Uses `THREE.InstancedMesh` per material palette, updating instance transformation matrices dynamically on block place/destroy.
 - Eliminates per-cube draw call overhead, ensuring buttery 60 FPS in iPad Safari.
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13. GLOBAL MASTER PRESET SYSTEM & MENU BUTTON ARCHITECTURE (ews.master/1)

30. Button Architecture & Layout Census:

- **Fixed Glass Controls (18 buttons):** Top-left mode stack (**SELECT**, **EDIT**, **BUILD**, **ANIMATE**), left camera/shading accordion (9 view & shading toggles), bottom simulation orbs (Pause, Speed, Cinematic).
- **Revolver 6-Shooter (6 radial buttons):** Movable radial touch dial serving as the primary hub (**Rig**, **Cutaway**, **Fluids**, **Contraband**, **Genetics**, **Settings**).
- **Sliding Sheets (62 contextual controls):** On-demand sliding inspection sheets adhering strictly to the Zero-Bar Layout Law, organized by domain.

31. Global Master Preset Engine (**copyMasterPacket** & **injectMasterPacket**):

- **Unified Schema (ews.master/1):** Bundles all sub-systems into a single coherent JSON payload:
 - **rig:** 23 canonical node positions, radii, scales, Horn, Wings, Mane style, and Mane visibility.
 - **viscera:** Stomach volume, cecum gas pressure, bladder volume, reproductive state, trimester, and organ bag capacities.
 - **world:** Active biome, 200x200 grid bounds, active Z-level, and painted mine/stockpile tile counts.
 - **camera:** Active camera mode (Chase, Orbit, Iso) and orientation angles.
 - **ui:** Active engine state and active sheet.
 - **Local Storage:** **saveGlobalPreset(name)** and **loadGlobalPreset(name)** persist complete game configurations locally across reloads.
 - **AI Bridge:** 1-tap **Export Master File for AI** copies the full state to the iPad clipboard; **Ingest AI Master File** applies AI-generated adjustments live in-place.
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14. REMOVAL OF ST3D IFRAME & UNIFIED IN-ENGINE 3D MODELER (ONE UNIVERSAL PONY)

32. The Root Cause of the "Two Models" & "Missing Revolver" Bug:

- Previous builds contained a legacy 54KB base64 script (**ST3D BUILD MODULE**) that created a duplicate hammer orb (**#bld**) at the bottom left. When clicked, it spawned a fullscreen iframe (**#st3dframe**, z-index 60) containing an isolated SDF raymarching editor.
- This iframe obscured the live 3D game, hid the Revolver (z-index 30), and displayed a disconnected static blockout mesh with no mane, tail, organs, horn, or wings.

33. The Unified In-Engine Architecture:

- **Zero Iframe:** The standalone iframe hack has been completely removed.
- **One Universal Pony:** All editing, modeling, building, and inspection occur directly on the single canonical 23-node Three.js character in the live scene.
- **Revolver Persistence:** The Revolver dial floats at z-index 30 and follows the user across all modes (**SELECT**, **EDIT**, **BUILD**, **ANIMATE**, **LIVE**).

34. Mode-Driven Game States (No Sub-Menus):

-  **SELECT Mode:** Tapping any node on the character highlights it with a 3D green selection halo, opens the node inspector, and allows copying node math without triggering walk locomotion.
-  **EDIT Mode (In-Engine Modeler):**
 - Renders a 3D selection halo and RGB axis lines (X red, Y green, Z blue) directly on the active node.
 - Opens the sliding **Node Inspector Sheet** with real-time sliders for Radius (r), Position (X, Y, Z), and Scale (sx, sy, sz).
 - Automatic L/R symmetry mirroring updates counterpart limbs in real-time.
 - Body connector cylinders, hair, and organs update continuously in the live scene as sliders move.
-  **BUILD Mode:**
 - Opens the **Colony Build & Excavation Menu** overlay for paintable mining (amber), stockpiling (cyan), and block placement.
 - Features a direct  **Open Node Modeler** button to transition smoothly to character editing.
-  **ANIMATE Mode:** Opens the procedural animation scrubber and module staging sandbox (Gas Room).

15. SYSTEM REDUNDANCY, DUPLICATION AUDIT & UNIFIED ENGINE INTEGRATION LAW

35. The Root Cause of Engine Duplication (The "Additive Patch" Anti-Pattern):

- Historically, when new features or fixes were added, previous AI turns often injected standalone modules or duplicate overlays (e.g. the 54KB ST3D iframe, duplicate hammer buttons, separate animation loops) rather than extending the core engine.
- This created parallel systems that fought over the same DOM and meshes, resulting in lost UI, missing hair/organs, and desynchronized game states.

36. Unified Engine Integration Law:

- Every new feature, editor tool, or behavioral stage **MUST** directly extend the core Three.js scene, the canonical 23-node character rig, and the active **GameMaster** state machine.
- Standalone iframes, duplicate viewport canvases, and parallel character models are strictly forbidden.

37. Key Redundancies Audited & Streamlined:

- **Gait System Clash:** Eliminated legacy primitive rotation swing in `updateWalking()`, delegating all leg articulation exclusively to `EWS_LEG_GAIT_v2` (authentic 4-beat lateral horse walk with folding joints).
- **Hidden Authored Assets Exposed:** The 4 authored hairstyles (`swept`, `scoot`, `crest`, `bob`) and 3 head morph presets (`G5 Chibi`, `G4 Classic`, `Wasteland Feral`) had no UI controls; now exposed directly in the **3D Avatar, Breeds & Features** sheet.
- **Camera Controls Unified:** Centralized camera modes into the  **Global Presets, Settings & AI Bridge** sheet and left accordion () , eliminating scattered duplicates.
- **Dissection Tools Consolidated:** Visceral cutaway slice, X-Ray ghosting, and layer isolation now share unified clipping state rather than competing for material visibility.

16. THE PRE-FLIGHT RIG & MATH SCRUTINY GATE (MANDATORY AI VERIFICATION PROTOCOL)

38. The Core Directive:

- Whenever any AI assistant (Gemini, Claude, ChatGPT) or human developer modifies code, generates an AI delta patch, or updates character geometry, they **MUST** execute the Pre-Flight Scrutiny Gate (`python3 tools/rig_math_gate.py`) and verify that all 5 verification checks pass before delivering files to Dylan.

39. The 5-Point Scrutiny Protocol:

- **Check 1: Single Scene Architecture (Zero Rogue Iframes):**
 - Verify zero fullscreen popups or iframes (`st3dframe` or equivalent).
 - All editing, modeling, building, and staging must exist natively in the primary Three.js viewport.
- **Check 2: Mathematical Rig Integrity & Symmetry:**
 - Validate all 23 canonical node positions (X forward, Y vertical, Z lateral) and radii ($r > 0$).
 - Ensure zero NaN, null, or unbounded infinity values.
 - Strict Bilateral Symmetry: Left and right limb pairs (FL1/FR1, BL0/BR0, cheekL/cheekR) must match within 0.02m on X and Y, with strictly mirrored Z coordinates ($Z_{\text{right}} = -Z_{\text{left}}$).
- **Check 3: Visceral Organ Bag & Volume Constraints:**
 - Centroid must anchor at midpoint: $C_{\text{bag}} = \frac{1}{2}(P_{\text{chest}} + P_{\text{rear}})$.
 - Resting capacity 65.0L, dynamic shrink floor 45.0L, maximum distension 120.0L.
 - Swell distension must scale only the ventral belly mesh downward/outward without clipping into forelegs.
- **Check 4: Universal Schema & AI Math Bridge Tagging:**

- Packets must declare valid schema headers: "schema": "ews.master/1" or "schema": "ews.math/1".
- Standardized coordinate units: "units": "1 = 1.0m; x fwd, y up, z lateral".
- Validated serialization and deserialization via copyMasterPacket() and injectMasterPacket().
- **Check 5: Zero-Bar Mobile UI Law:**
 - Touch targets must be $\geq 44\text{px}$ (frosted glass orbs at 48px, controls at 44px).
 - Exactly zero persistent top or bottom chrome bars; 100% borderless 3D viewport.
 - Revolver 6-shooter dial must maintain highest z-index layer (z-index: 30) and remain draggable across all engine states.

17. DYNAMIC NODE MAPPING GUIDE & MOD/STORY SOCKET SCHEMA (EWS-MAP/1)

40. Why the Mapping Guide Prevents Engine & Mod Collapse:

- As Dylan models in the 3D editor, nodes may be renamed, added (e.g. wings, horns, ears, gryphon parts), or moved. If tools or story scripts rely on hardcoded node names, the entire system breaks whenever the anatomy evolves.
- The **Universal Node Map** ([Rigs_and_Art/node_map.json](#)) acts as the engine's semantic Rosetta Stone. Story sequences, visual editor scenes, clothing mods, and combat hitboxes reference **Sockets** and **Roles** (e.g., `socket_hat` or `role: "root"`), never volatile raw indices.

41. Socket & Mod Attachment Mapping:

- `socket_hat`: Anchors to `head` (helmets, caps, goggles, eyewear).
- `socket_horn`: Anchors to `head` with forward tilt angle (magic horns, battle armor).
- `socket_crest_mane` & `socket_forelock`: Anchors along the dorsal ridge and brow.
- `socket_tail`: Anchors to `pelvis` dock.
- `socket_saddle` & `socket_barding_chest`: Anchors to `chest` (saddles, packs, harnesses).
- `socket_left_wing` & `socket_right_wing`: Anchors to `chest` flanks.
- `socket_hoof_FL/FR/BL/BR`: Anchors to end-effectors (FL4, FR4, BL4, BR4) for horseshoes, combat wraps, and tools.
- `socket_organ_bag`: Anchored at $C_{\text{bag}} = \frac{1}{2}(P_{\text{chest}} + P_{\text{rear}})$.

42. Dynamic Scrutiny Gate Integration:

- The Pre-Flight Scrutiny Gate ([tools/rig_math_gate.py](#)) dynamically reads `node_map.json`.

- When new nodes or species (e.g. Gryphons, Changelings) are authored, the gate validates the tree structure, connectivity, and bilateral symmetry against the schema rather than rejecting unfamiliar node names.
-

18. GAMEMASTER ENTITY-COMPONENT ARCHITECTURE & PLUGGABLE ACTOR SCHEMAS (EWS-ACTOR/1)

43. Decoupled 5-Layer Component Model:

- Every actor in EWS (player pony, settler, feral scout, automaton robot) is assembled from 5 decoupled, pluggable component layers:
 - **Identity & Genetics:** Unique ID (`ACTOR:EQUINE:*`), Name, Species, Breed, Mendelian alleles (`$E/e, A/a, Cr/n$`).
 - **S.P.E.C.I.A.L. & Combat:** Strength, Perception, Endurance, Charisma, Intelligence, Agility, Luck, Max HP, Action Points, and Carry Weight.
 - **Mind & Behavioral Logic (Pluggable):** Can be `biological_personality` (traits, potty-training progression, morality) OR `robot_core_logic` (directives, threat matrix, battery %, coolant temp).
 - **Visceral System (Pluggable):** Can be `hindgut_fermentation` (stomach, cecum gas, bladder, organ bag capacity) OR `synthetic_hydraulics` (coolant reservoir, PSI, fusion cell).
 - **3D Rig & Socket Binding:** Binds to `Rigs_and_Art/node_map.json` (`socket_hat`, `socket_saddle`, `socket_horn`, `socket_left_wing`, `socket_hoof`).

44. Why Swapping Modules (e.g. Robot Core Logic) Never Collapses the Engine:

- The engine operates on interface contracts and semantic tags, not rigid monolithic classes.
- If a mod or story designer replaces biological personality with Robot Core Logic, the 3D physics rig, inventory, and combat formulas continue functioning without errors. Systems query component tags (`hasComponent('mind')`, `actor.viscera.type`) to dynamically execute the appropriate behaviors.

45. Universal Catalog Taxonomy (EWS-GM-CATALOG/1):

- Items: `ITEM:WEAPON:*`, `ITEM:ARMOR:*`, `ITEM:CONSUMABLE:*`, `ITEM:CONTRABAND:*`.
 - All items equip into named sockets defined in `node_map.json`, ensuring clothing, weapons, and accessories adapt to any modified character geometry automatically.
-

19. BUILD 7 INTEGRATION: MULTI-RING REVOLVER HUD, 3D GIZMOS & ORGANIC SKINNING ROADMAP

46. Build 7 Upgrades Overview (EWS_with_Build_7.html):

- **Multi-Ring Dual-Row Revolver:**
 - Layer 1 (Inner Ring: 36px bullets, $R_{inner} \geq 46px$).
 - Layer 2 (Outer Ring: 32px bullets, $R_{outer} = R_{inner} + 36px$ kissing clearance).
 - Layer 3 (Folder Expansion Ring: 30px bullets).
 - Strict chord-distance formula $C = 2 R \sin(\pi / N)$ prevents overlapping icons.
 - Live Geometry Sliders in Settings (⚙️): Overall Scale (50%-150%), Ring Gap (0-24px), Bullet Size (16-44px).
 - High-contrast uppercase text labels under icons.
 - Centered coordinate positioning ensuring in-bounds spawning on all viewports.
- **Interactive 3D Transform Gizmos (No Sliders in Build):**
 - Translation Gizmo: 3-axis RGB arrows (X red forward, Y green up, Z blue lateral) + yellow center grab.
 - Rotation Gizmo: 3-axis Torus rings.
 - Scale Gizmo: 3-axis stretch box handles + center uniform scale cube.
 - Drag planes constructed dynamically orthogonal to camera ray; 60-step undo/redo stack (`_pushNodeUndo`).
- **Organic Surface Contour Outline:**
 - Form-fitting organic selection outline (THREE.BackSide scale expansion) without boxy bounding boxes.
 - Sleek skeletal joint rig with small yellow nodes ($r = 0.016$) and thin green bone connection lines.
- **Modeler Isolation (MODEL):**
 - Clean staging with grid, camera locked to pony, pause state preserved, exit door (🚪) returning to Build.

47. Roadmap: Seamless Organic Skinning & Mesh Blending:

- **Root Cause of Joint Creases:** Independent intersecting THREE.Mesh objects have discontinuous vertex normals at seams, producing visible lighting creases.
- **Path 1 (Recommended for Mobile - 60 FPS): Tangent Lofts with Vertex Normal Welding:**
 - Parametric Hermite/Bézier cylinders bridging parent/child spheres with averaged shared normals across boundary rings.
- **Path 2: Signed Distance Field (SDF) + Real-Time Marching Cubes:**
 - Evaluates analytical smooth minimum (smin) field to extract a single continuous polygon mesh.

20. SEAMLESS TANGENT LOFT IMPLEMENTATION & COMPREHENSIVE ROADMAP AUDIT

48. Option A Implementation (Zero Joint Seams):

- **Welded Tangent Lofts (`createWeldedLoftGeometry`):** Replaced raw flat cylinders in `initBodyConnectors()` with procedural Hermite tangent loft geometries.
- **Continuous Surface Normal Welding:**
 - Vertices at the top parent junction tilt upward (N_{top}) to match the parent sphere dome.
 - Vertices at the bottom child junction tilt downward (N_{bottom}) to match the child sphere dome.
 - Smooth continuous cosine interpolation across the shaft eliminates hard lighting breaks and dark crease lines.
 - Limbs, neck, and torso render as a continuous, unified piece of organic anatomy while preserving 60 FPS mobile performance.

49. Master Audit of Deferred & Promised Features (The Lorebook Reminder Matrix):

- **Item 1: 3D Voxel World Grid & Active Terrain Rendering:**
 - *Status:* Math mapped in `GameMaster.js` (200x200 tiles, 3 biomes, stream spline, horizon mountain skirt); pending live 3D visual rendering in the viewport.
- **Item 2: Modeler Reference-Image Backdrop:**
 - *Status:* Requested in Addendum v2.1; allows dropping a 2D sketch or character turnaround onto a semi-transparent plane behind the pony in `EDIT` mode for tracing.
- **Item 3: Pose & Keyframe Clip Saver:**
 - *Status:* Keyframe animation format defined in `stabletec.blockout/2`; pending 1-tap pose snapshot button for custom animations.
- **Item 4: Story Editor & Visual Creation Sockets:**
 - *Status:* Semantic socket architecture completed in `Rigs_and_Art/node_map.json`; pending UI wiring in Story mode.
- **Item 5: Diapers & Foal Potty-Training Progression (Package O):**
 - *Status:* Fully documented in lore and GameMaster schema (Levels 0-2); diaper equipment visual attachment pending.

21. PROCEDURAL MATERIAL PHYSICS, WEARABLE DIAPER ABSORPTION & VISCERAL HYDROSTATICS

50. Lightweight PBR Material Physics (Pure Math, Zero Textures):

- Grounded in physical optical constants from [RefractiveIndex.INFO](#) and the [Pixel & Poly IOR Database](#):
 - Water: $n = 1.333$, Saline Urine: $n = 1.336$, Soil: $n = 1.480$, Obsidian: $n = 1.500$, Bone: $n = 1.550$, Cotton: $n = 1.530$.
- **Disney Sheen Cloth Math**: Evaluates grazing microfiber fuzz scattering using an inverted Fresnel curve ($f_{\text{sheen}} = (1 - \cos \theta)^5$) for velvet, denim, and burlap without downloading raster textures.
- **Oren-Nayar Earth Shading**: Computes micro-pit retro-reflection for dry wasteland dirt, mud, and geological sedimentary rock strata ($\sin(y \times 14.0 + \text{noise}(x,z) \times 2.8)$).
- **Subsurface Scattering Living Tissue**: Thin ear, flank, and organ mucosa evaluation using dipole transmission ($R(r)$) producing warm vascular transillumination.

51. The Diaper Wearable Item & Superabsorbent Polymer (SAP) Absorption

Mechanics:

- **Socket Anchoring**: Binds to **pelvis** (sacral dock), **rear** (haunches), and **BL0/BR0** (inguinal thigh gathers).
- **Multi-Layer Shader Architecture (60 FPS Mobile)**: Utilizes a single dynamic 256×256 canvas composite texture driving **material.map** and **material.roughnessMap**:
 - *Layer 1 (Outer Backing)*: Breathable waterproof polyethylene film (roughness 0.45).
 - *Layer 2 (Core Matrix)*: Porous fibrous cellulose fluff matrix.
 - *Layer 3 (Wetness Indicator)*: Centerline stripe shifting from dry golden-amber (**#fad165**) to vivid blue (**#00b4d8**) at $\geq 25 \text{ mL}$ fluid impact.
 - *Layer 4 (Dampness Bloom)*: Dynamic radial dampening darkening the diffuse color and dropping roughness if saturated $> 100\%$ ($> 650 \text{ mL}$).
- **Real-World Hydrogel Swelling Math**:
 - Volumetric Core Swell: $V_{\text{core}} = V_{\text{dry}} + 0.95 \times V_{\text{absorbed}}$.
 - Gravitational Crotch Sag: $\Delta y = -0.00012 \times m_{\text{absorbed_grams}}$ (up to 6.5 cm downward sag).
 - Lateral Thigh Bulge: $W_{\text{crotch}} = W_0 \times \left(1.0 + 0.48 \sqrt{V_{\text{absorbed}} / V_{\text{max}}}\right)$ (up to 145% lateral width expansion).
- **Fluid Engine Coupling**: Voiding the bladder transfers volume directly from **ORGANS.bladderVol** into **diaper.absorbedVol**. Prevents ground puddle splats while capacity remains under 650 mL .

52. Visceral Organ Bag Hydrostatic Coupling (The One True Source):

- Centroid maintains $C_{\text{bag}} = \frac{1}{2}(P_{\text{chest}} + P_{\text{rear}})$.

- Dynamic volume tracking ($45^\circ \text{L} \rightarrow 65^\circ \text{L}$ rest $\rightarrow 120^\circ \text{L}$ max distension).
- 3D Inversion Spill: When orientation inverts ($\mathbf{u} \cdot \hat{\mathbf{y}} < -0.40$), gravitational head pressure exceeds esophageal threshold, venting fluid boluses through the oral aperture.